

code

effect

`x, y = next(loader)`

fresh shuffled batch

`logits = model(x)`

records graph + activations

`loss = loss_fn(logits, y)`

raw logits in, scalar out

`opt.zero_grad()`

$p.grad \leftarrow \text{None}$

`loss.backward()`

$p.grad \leftarrow \nabla L$ , graph freed

`clip_grad_norm_(ps, c)`

rescale if  $\|g\| > c$ , return  $\|g\|$

`opt.step()`

update  $p$  from  $p.grad + \text{state}$

`sched.step()`

set next LR  $\eta$

`log(loss, gnorm, lr)`

unclipped norm, current LR

next step